

DP Computer Science: A 4.4.1 / A 4.4.2 Ethics of Emerging Technologies

1. Core Concepts & Learning Objectives

Learning Intentions

By the end of this lesson, students will be able to:

- Discuss a wide range of ethical issues in ML, including **accountability, fairness, bias, privacy, and transparency**
 - Analyse the ethical challenges of ML in specific contexts, such as **online communication** (e.g., misinformation, harassment)
 - Discuss the potential **societal implications** of emerging technologies like quantum computing, AR, and VR
 - Explain why **ethical guidelines must be continually reassessed** as technology advances
 - Make **conceptual links** between machine learning and other areas of the computer science syllabus
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Key Vocabulary

Term	Definition
Accountability	The responsibility for the outcomes of an AI system; who is held responsible when things go wrong
Fairness	Ensuring AI systems do not discriminate against individuals or groups
Bias	Systematic error in a model that leads to unfair or discriminatory outcomes
Privacy	The protection of personal data and the right to control how it is collected and used

Term	Definition
Transparency	The degree to which an AI system's decision-making process is open and understandable
Consent	The permission given by individuals for their data to be collected and used
Misinformation	False or misleading information spread online, often amplified by algorithms
Environmental impact	The energy consumption and carbon footprint of training large AI models
Algorithmic justice	The fair and equitable application of algorithms in society
Ethical guidelines	Principles and standards that govern the responsible development and use of technology

Approaches to Learning (ATL) Elements

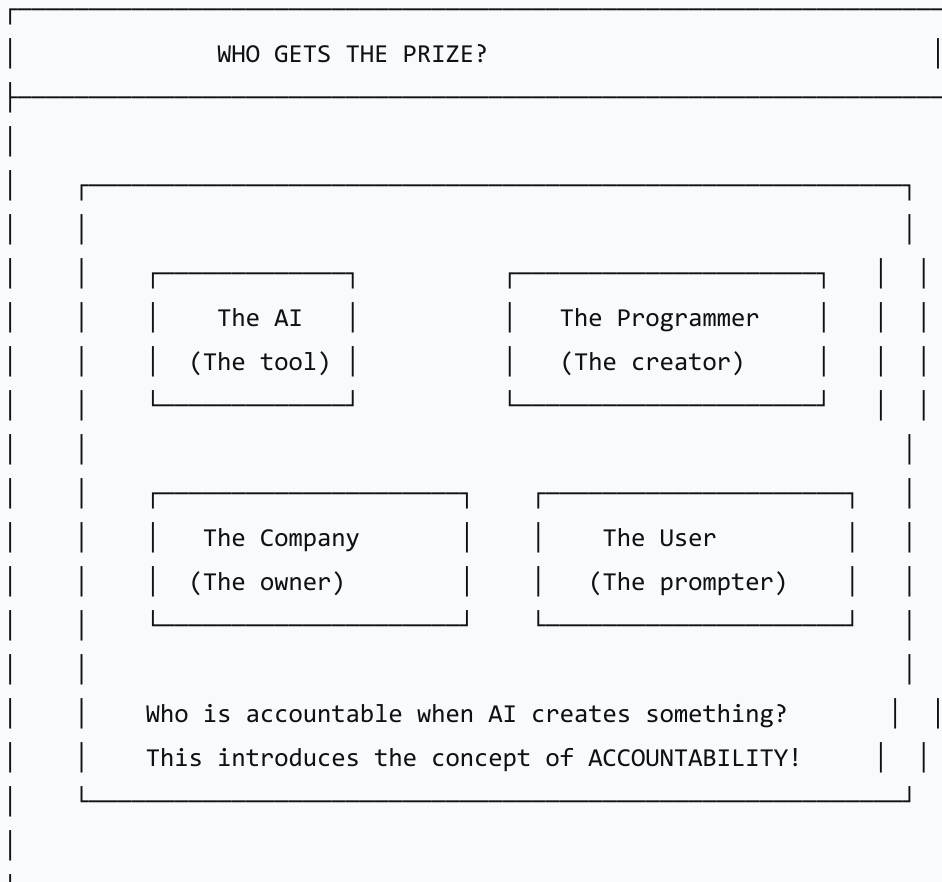
Skill	Description
Thinking (Ethical Reasoning & Critical Thinking)	Evaluating complex ethical dilemmas; analysing the long-term societal impact of technology
Communication (Collaboration & Discussion)	Collaborating effectively in group discussions to explore multifaceted ethical issues and articulate reasoned arguments
TOK Links	Questioning the responsibilities of knowledge creators, the nature of algorithmic truth, and the ethical constraints on technological advancement

2. Introduction: The Hook

The Provocative Scenario

"An AI artist creates a prize-winning painting. Who gets the prize money? The AI? The programmer? The company that owns the AI?"

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3. The Ethical Minefield of Machine Learning

Key Ethical Issues in ML

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ACCOUNTABILITY

- Who is responsible when an AI fails?
- The developer? The user? The company?
- Self-driving car accidents, medical misdiagnosis

FAIRNESS & BIAS

- Bias in training data → bias in outcomes
- Discriminatory loan approvals, hiring decisions
- "Garbage In, Garbage Out" but with real people

PRIVACY & CONSENT

- Who owns the data?
- Was consent obtained?
- Data collection, surveillance, tracking

TRANSPARENCY

- Can we understand how the model makes decisions?
- "Black box" vs. "White box" models
- Right to explanation

ENVIRONMENTAL IMPACT

- Energy consumption of training large models
- Carbon footprint of AI
- Sustainability concerns

Ethics in Online Communication

text

ETHICS IN ONLINE COMMUNICATION

MISINFORMATION

- Algorithms amplify false content
- Echo chambers and filter bubbles
- Impact on elections, public health

HARASSMENT & TOXICITY

- Algorithms can amplify toxic content
- Lack of accountability for online behaviour
- Moderation challenges

BIAS IN NEWS FEEDS

- Algorithms curate what we see
- Can reinforce existing biases
- Can create polarisation

4. Activity: "Future Shock" Scenarios

Instructions

1. In groups, you are assigned one **emerging technology**:

- Quantum Computing
- Augmented Reality (AR)
- Virtual Reality (VR)

2. For your assigned technology, brainstorm:

- One positive societal impact
- One negative ethical implication

Quantum Computing

Aspect	Description
Positive Impact	Revolutionise drug discovery, materials science, cryptography
Ethical Concern	Could break current encryption, threatening security, privacy, and online banking

Augmented Reality (AR)

Aspect	Description
Positive Impact	Enhanced education, training, and navigation
Ethical Concern	Privacy (constant surveillance via cameras), manipulation of reality

Virtual Reality (VR)

Aspect	Description
Positive Impact	Immersive education, therapy, and remote collaboration
Ethical Concern	Addiction, disconnection from reality, psychological harm

5. Why Ethical Guidelines Must Evolve

Technology vs. Ethics

text

WHY ETHICAL GUIDELINES MUST EVOLVE

TECHNOLOGY MOVES FAST

- New capabilities emerge before we understand their implications
- Laws and guidelines lag behind

NEW UNFORESEEN SCENARIOS

- Deepfakes didn't exist 5 years ago
- AI art creation raises new copyright issues
- LLMs can generate convincing misinformation

GLOBAL IMPACT

- Technology affects different communities differently
- One-size-fits-all guidelines may not work

SOCIETAL VALUES CHANGE

- What is acceptable today may not be acceptable tomorrow
- Ethical standards evolve with society



Examples of Evolving Guidelines

Technology	Old Concern	New/Emerging Concern
Social Media	Privacy	Misinformation, polarisation, mental health
AI	Automation	Bias, accountability, deepfakes
Data	Storage	Surveillance, consent, data ethics
Quantum Computing	Theoretical	Breaking encryption, security threats

6. Linking Questions: Connecting ML to Other CS Topics

Sample Linking Questions

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LINKING QUESTIONS

NETWORKS (A2)

How could ML be used to detect network intrusions?

How might quantum computing affect network security?

DATABASES (A3)

How could ML improve database query optimisation?

What are the privacy implications of using ML on databases with personal data?

SYSTEM FUNDAMENTALS (A1)

How could ML be used to optimise CPU scheduling?

What are the hardware requirements for training ML models?

OBJECT-ORIENTED PROGRAMMING (A5)

How could you design an OOP system for an ML pipeline?

How do inheritance and polymorphism apply to ML models?

7. Lesson Sequence

Lesson Plan (60 minutes)

Phase	Time	Activity
Recap & Hook	10 min	Recap model selection ethics; AI artist scenario (accountability)
Lecture: Ethical Issues	15 min	Accountability, fairness, bias, privacy, transparency, environmental impact
Group Activity	15 min	"Future Shock" — students analyse positive/negative impacts of emerging technologies
Discussion	15 min	Groups share findings; discuss why ethical guidelines must evolve; linking questions

Phase	Time	Activity
Consolidation	5 min	Recap; preview next lesson

Discussion Questions

Question	Purpose
"Who is accountable when an AI makes a mistake?"	Understand accountability
"How can bias enter an ML system?"	Understand bias
"Why is transparency important in AI?"	Understand transparency
"What are the environmental costs of AI?"	Understand sustainability
"Why must ethical guidelines evolve?"	Understand evolving ethics

8. Assessment Activities

Assessment Type	Description
Class Participation	Observe engagement and contributions
"Future Shock" Activity	Assess ability to analyse positive/negative impacts
Exit Ticket	Gauge understanding and identify areas needing clarification

Sample Exit Ticket Questions:

1. Give one ethical issue related to machine learning.
2. Why is bias a concern in AI systems?
3. What is a potential positive and negative impact of quantum computing?
4. Why must ethical guidelines be continually updated?

9. Differentiation

Level	Strategy
Support	Provide structured frameworks for ethical analysis; use current, relatable examples
Challenge	Research specific cases of algorithmic bias; explore international ethical guidelines

10. Resources

- **Presentation** (Slide Deck)
 - **eBook** (A4.4.1 & A4.4.2 with examples)
 - **Worksheet:** Ethics scenarios and linking questions
 - **Worksheet Answers**
 - **Quizlet** (A4.4.1/A4.4.2 vocabulary flashcards)
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11. Summary: Key Takeaways

Concept	Key Point
Accountability	Who is responsible when AI fails?
Bias	Training data bias → outcome bias
Privacy	Data collection and consent are critical
Transparency	Models should be explainable
Environmental Impact	AI training is energy-intensive
Misinformation	Algorithms can amplify false content

Concept	Key Point
Emerging Tech	Quantum, AR, VR bring new ethical challenges
Evolving Guidelines	Ethics must keep up with technology

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KEY REMINDERS
✔ Accountability: Who is responsible? ✔ Bias: Data bias → model bias ✔ Privacy: Protect personal data ✔ Transparency: Explainable AI ✔ Environmental impact: AI is energy-intensive ✔ Misinformation: Algorithms can amplify false ✔ Emerging tech: New challenges ✔ Ethical guidelines must evolve

"Technology is not neutral—it reflects the values, biases, and choices of its creators. Ethical awareness is not an afterthought; it is a fundamental part of responsible innovation."